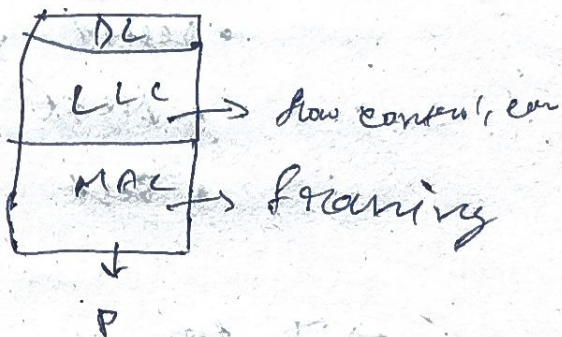
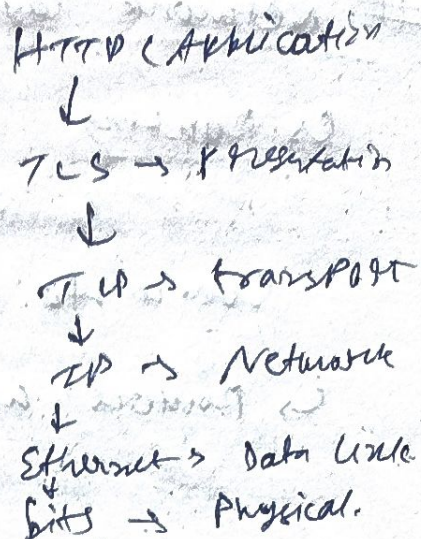
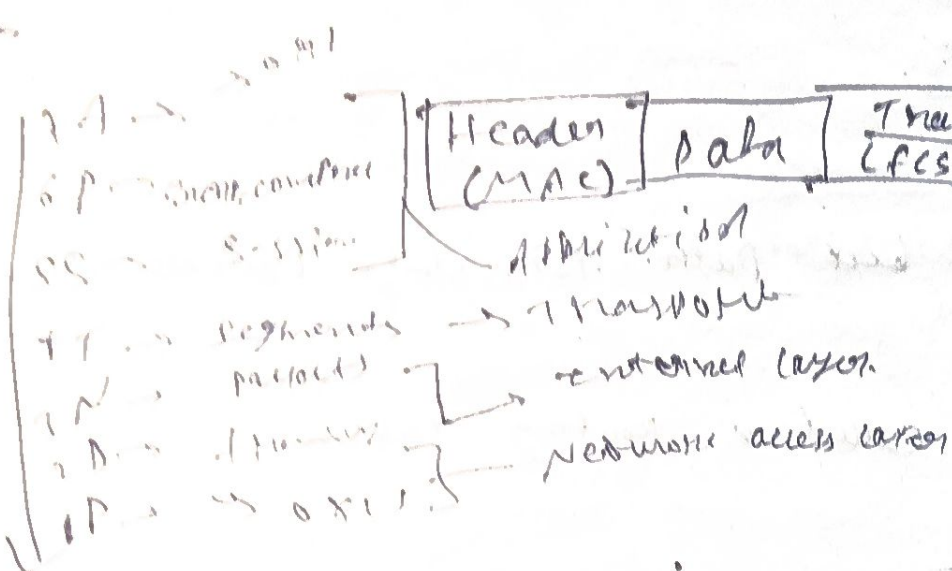


CIN mod-1



Mod-2

Network protocols:

1) A protocol is simply set of rules that devices follow to communicate.

2) TCP/IP protocols:

* Application layer:

uses interacts with protocols:

* Transport layer:

uses TCP

uses UDP

uses HTTP

uses FTP

uses SMTP

Internet layer

handles IP addressing & routing

* Network access layer:

uses Ethernet

uses MAC

ATTN

IPv4

class A

class B

C

1 - 126

128 - 191

192 - 223

large

medium

small

Module - 3

Network Layer: - IP Routing

↳ Routing means:

1 finding the best path for data packets from source to destination.

↳ Routers use something called a Routing Table.

* What is routing table:

↳ A routing table contains:

- Destination network
- Next hop
- Best Path

Example:

<u>Destination</u>	<u>Next Hop</u>
192.168.1.0	Router A
10.0.0.0	Router B

→ Router check this table before forwarding packets.

* Type of Routing:

1) Static Routing.

2) Dynamic Routing.

1) Static Routing:

↳ Static routing is also called non-adaptive routing.

Meaning:

↳ Routes are manually configured by the network administrator.

↳ Routers does not change automatically routes.

If network changes → admin must manually update routes.

* How it works:

Admin manually enters:

- Destination IP
- Subnet mask
- Next hop IP

Adv of static routing:

- ↳ more secure:
 - NO automatic route sharing
 - ↳ low Bandwidth usage:
 - Routers don't exchange routing updates.
 - ↳ NO CPU overhead:
 - NO complex calculations
 - ↳ simple for small networks
- good for:
- small offices
 - simple LANS

2) Dynamic routing:

↳ It is also known as Adaptive routing.

Routers automatically:

- learn routes
- update routing tables
- find best path

using routing protocols.

How it works:

→ Routers exchange information with neighbouring routers.

They use:

- Algorithms
- Metrics
- Routing Protocols

To determine best path

Common Dynamic routing protocols:

- ↳ RIP
- ↳ OSPF
- ↳ BGP

Disadv of static routing

- X Difficulty in large network
 - ↳ what if configuring 300 routers manually.
- X No automatic recovery
 - if one route fails:
 - ↳ routers can't find alternate path automatically.
- X Time consuming
 - ↳ manual configuration required.

* Routing Metrics:

↳ Routers decide best path using metrics like:

<u>Metric</u>	<u>meaning</u>
Hop count	no. of routers crossed
bandwidth	speed of link
Delay	time taken
Cost	network expense

* Adv. of Dynamic routing:

- ↳ Automatic updates
- ↳ Best and efficient path selection
- ↳ Handles failures automatically.
- ↳ If one route fails switch/select to another route

* Disadv

- more CPU usage
 - ↳ Routers continuously calculate routes
- more Bandwidth consumption
 - ↳ Routers exchange information regularly.
- less secure.

Routing Protocols:

* RIP:

↳ RIP is a distance-vector routing protocol used in dynamic routing that determines the best path based on hop count.

"Distance" = how far (no. of routers/hop)

"Vector" = direction (which path to take)

So RIP decides:

which path has the least number of routers.

* Working of RIP:

- 1) Each router maintain a routing table
- 2) It shares this table with neighbours every 30 seconds.
- 3) Neighbour routers update their tables.
- 4) Best path = min^m hop count.

* Metric used in RIP

→ only hop count

- 1 hop = 1 router
- max hop allowed = 15.

* what happen after 15 hops?

↳ 16 hops = infinite (unreachable)

* Key features:

- ↳ simple protocol, easy to configure
- ↳ uses UDP, port number: 520.
- ↳ period updates: every 30 seconds
- ↳ uses bellman ford algorithm.

Types of RIP

RIP Version 1 (RIPv1)

- ↳ classful
- ↳ Does not support Subnetting
- ↳ No authentication

* Advantages:

- ↳ Easy to implement
- ↳ low complexity
- ↳ Suitable for small networks

RIP Version 2 (RIPv2)

- ↳ classless
- ↳ support CIDR & Subnetting
- ↳ support authⁿ.

* Disadvantages:

- ↳ slow convergence: takes time to update network changes
- ↳ limited to 15 hops
- ↳ Routing loop problem: packets may circulate endlessly.

Does not consider:
↳ Not efficient: bandwidth, delay, load

* Techniques to solve RIP Problems:

- ↳ split horizon: Don't send route info back to where it came from
- ↳ Poison Reverse: send route as unreachable to prevent loop
- ↳ hold down timer: wait before accepting new updates.

* OSPF:

↳ OSPF is a link-state routing protocol that find the best path using shortest path algorithm (Dijkstra) based on Cost (bandwidth).

* link-state:

- Each router builds a complete map of the network
- then calculates the best path using algo.

* working of OSPF:

- 1.) Neighbour Discovery: Routers find nearby routers
- 2.) Exchange info: link states (network info)
- 3.) Build topology DB: Each routers creates full new map
- 4.) Apply algorithm: use Dijkstra's Algo.
- 5.) Best path select: Shortest (lowest cost) path is chosen.

* Features of OSPF:

- ↳ uses Dijkstra's algo.
- ↳ fast convergence
- ↳ support large networks
- ↳ uses cost
- ↳ classes (support CIDR, subnetting)

* OSPF uses:

- ↳ Protocol: TCP
- ↳ Port: 89

* Adv of OSPF:

- ↳ faster than RIP
- ↳ No hop limit
- ↳ Better path selection
- ↳ scalable (large net)
- ↳ loop-free (almost)

* Disadv:

- ↳ complex to configure
- ↳ high CPU & memory usage
- ↳ Requires Planning (areas)

→ OSPF divides network into areas:

Why?

- ↳ Reduce complexity
- ↳ improve performance

Types:

- Backbone Area (area 0) → main area
- Other areas connect to backbone.

→ All OSPF areas must connect to area 0.

* BGP:

↳ BGP is a Path-Vector routing protocol used to exchange routing information b/w different net (Autonomous Systems) on the internet.

↳ An autonomous system (AS) is a group of net under one organization with a common routing policy.

ex:

- ↳ Airtel network
- ↳ sid
- ↳ google

} → Each has a unique AS number.
ASN

* Core ideas of BGP:

- RIP → hop count
- OSPF → cost
- BGP → path (AS path)

↳ BGP choose route based on best path b/w net.

* What is Path Vector:

- BGP stores the entire path (list of AS numbers)
- Help avoid loops and select best route.

Types of BGP:

1) eBGP (external)

- ↳ B/W different AS
- ↳ used on internet

2) iBGP (internal)

- ↳ within in same AS
- ↳ used inside organization

* working of BGP:

- 1) Routers establish connection (Peer routers)
- 2) exchanging routing info
- 3) store path in ~~the~~ routing table.
- 4) select best path based on rules.

* How BGP select best path:

↳ BGP uses multiple factors:

- AS path length
- Next hop
- Policies
- New rules

* Features:

- ↳ Path vector protocol
- ↳ Policy-based routing
- ↳ highly scalable.
- ↳ used in internet backbone
- ↳ Prevents loop using AS path

Protocol details

- ↳ works on TCP
- ↳ Port: 179

* Advantages of BGP

- ↳ controls routing policies
- ↳ works b/w large n/w
- ↳ Scalable for internet
- ↳ loop prevention (As Path)

* Disadv:

- ↳ very complex
- ↳ slow convergence
- ↳ Require high configuration knowledge.

RIP → small n/w

OSPF → big internal n/w

BGP → Internet-level routing

BGP is used to route data b/w different n/w on the internet using path-based decision making.